



AI Realm Implementation

Technical Specification

Contents

1	Executive summary	4
2	Agentic Solution Overview	5
2.1	Agentic Solution Design.....	5
2.2	Step-by-Step Component Breakdown	5
2.3	Flow Summary	8
3	AWS Solution Architecture Diagram.....	9
3.1	Component Descriptions.....	10
3.1.1	Identity & Access	10
3.1.2	Frontend Layer (ECS Fargate + ALB).....	10
3.1.3	Backend Layer (ECS Fargate + ALB).....	10
3.1.4	Networking.....	10
3.1.5	AI & Knowledge Layer.....	10
3.1.6	Data Management	11
3.1.7	Data Integration.....	11
3.1.8	Observability & Security.....	11
4	Component Choices.....	13
5	Authentication/Authorization	14
6	High availability/Disaster Recovery.....	14
7	Data Lifecycle.....	14
8	Bedrock.....	15
8.1	Embedding Models.....	15
8.2	Large Language Models (LLMs)	15
8.3	Amazon Bedrock Guardrails.....	16
9	Tech Stack & CI/CD Strategy.....	17
10	Non-Functional Requirements (NFRs)	18
11	AWS Cost Assessment.....	19
12	AZURE Solution Architecture Diagram.....	20
12.1	Component Description	21
12.1.1	Identity & Access	21
12.1.2	Frontend Layer	21
12.1.3	Backend Layer	21
12.1.4	Networking.....	21
12.1.5	AI & Knowledge Layer.....	21
12.1.6	Data Management	22
12.1.7	Observability & Security.....	22

12.2	Old Mutual Availability of Azure Services	23
13	Azure Architecture Solution Estimated Pricing	Error! Bookmark not defined.

AI Realm Implementation

1 Executive summary

This technical specification outlines Phase 1 of extending the Regi compliance agent into a broader multi-agent AI system (AI Realm) for Old Mutual's compliance function. The initiative will adapt iOCO's established multi-agent framework to operate seamlessly within Old Mutual's environment, ensuring compatibility with existing security, governance, and integration requirements.

The solution will evolve the current compliance agent into a scalable multi-agent ecosystem capable of supporting additional specialized agents. These agents will augment compliance operations by performing functions such as:

- **Regulatory monitoring** – continuous scanning of regulatory changes and obligations.
- **Documentation analysis** – intelligent processing of compliance documentation, policies, and procedures.
- **Risk flagging and early alerts** – proactive identification of potential compliance risks.

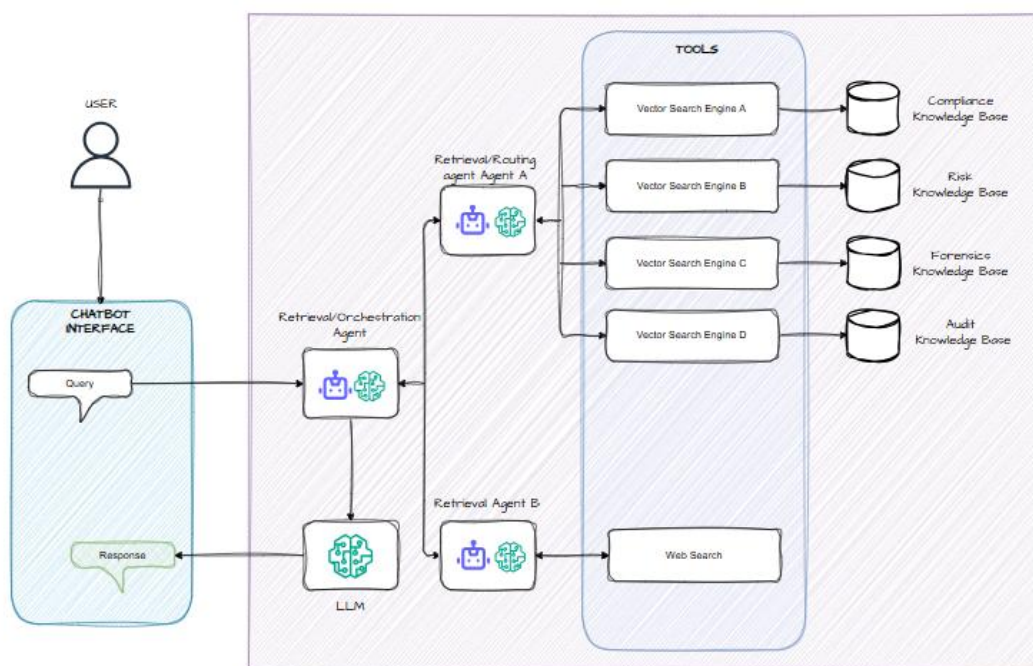
The deployment will emphasize security, auditability, and least-privilege access, aligned with Old Mutual's enterprise standards. This includes integration with Entra ID (SAML-based authentication), role-based authorization, and adherence to multi-AZ, serverless architecture patterns to support high availability and disaster recovery.

Upon completion of Phase 1, the AI Realm will be positioned to expand beyond compliance into actuarial, forensic, and risk analysis domains, establishing a foundation for a robust, AI-assisted compliance environment that improves efficiency, accuracy, and regulatory resilience.

2 Agentic Solution Overview

This architecture represents a multi-agent Retrieval-Augmented Generation (RAG) solution. It enables a user to query a chatbot, which then orchestrates specialized agents to search across multiple knowledge bases and external sources, before synthesizing the final answer with a Large Language Model (LLM).

2.1 Agentic Solution Design



2.2 Step-by-Step Component Breakdown

a). User

- The end-user initiates a query.
- Interaction is entirely through the chatbot interface.

b). Chatbot Interface

- **Purpose:**
Acts as the frontend UI where users type their questions and receive responses.
- **Function:**
Captures the query and sends it to the backend orchestration agent.
- **Fit in the solution:**
Isolates user interaction from backend logic so users never directly interact with internal systems.

c) Retrieval/Orchestration Agent

- **Purpose:**
Acts as the brain coordinating how queries are processed.
- **Function:**
 - Receives query from chatbot.
 - Decides whether to route it to:
 - **Retrieval/Routing Agent A** for internal knowledge bases.
 - **Retrieval Agent B** for web search.
 - Once results are fetched, forwards them to the LLM for synthesis.
- **Fit in the solution:**
This is the central traffic controller ensuring the right agent and tools are used for each query.

d) Retrieval/Routing Agent (Agent A)

- **Purpose:**
Specializes in retrieving structured knowledge from multiple internal knowledge bases.
- **Function:**
 - Breaks down the query to determine which Vector Search Engine(s) to query.
 - Can search one or multiple knowledge bases in parallel.
- **Fit in the solution:**
Ensures optimal, targeted retrieval rather than brute-force searching all databases.

e) Vector Search Engines (A, B, C, D)

- **Purpose:**
Perform semantic vector searches to find relevant documents based on meaning, not just keywords.
- **Function:**
 - Each engine is connected to a specific knowledge base:
 - A → Compliance KB
 - B → Risk KB
 - C → Forensics KB
 - D → Audit KB
 - Return ranked, relevant content to Agent A.
- **Fit in the solution:**
Enable **fast, accurate semantic retrieval** for domain-specific knowledge.

f) Retrieval Agent B

- **Purpose:**
Handles **external** information gathering.
- **Function:**
 - Runs queries against web search APIs.
 - Returns results for synthesis with internal KB data.
- **Fit in the solution:**
Allows the chatbot to answer **open-domain** or **non-local** questions.

g) LLM (Large Language Model)

- **Purpose:**
Synthesizes retrieved content into a **human-like, contextually relevant response**.
- **Function:**
 - Takes retrieved documents/snippets from Agent A and/or B.
 - Generates a coherent, natural-language answer.
- **Fit in the solution:**
Converts raw retrieved data into a **final polished response** for the user.

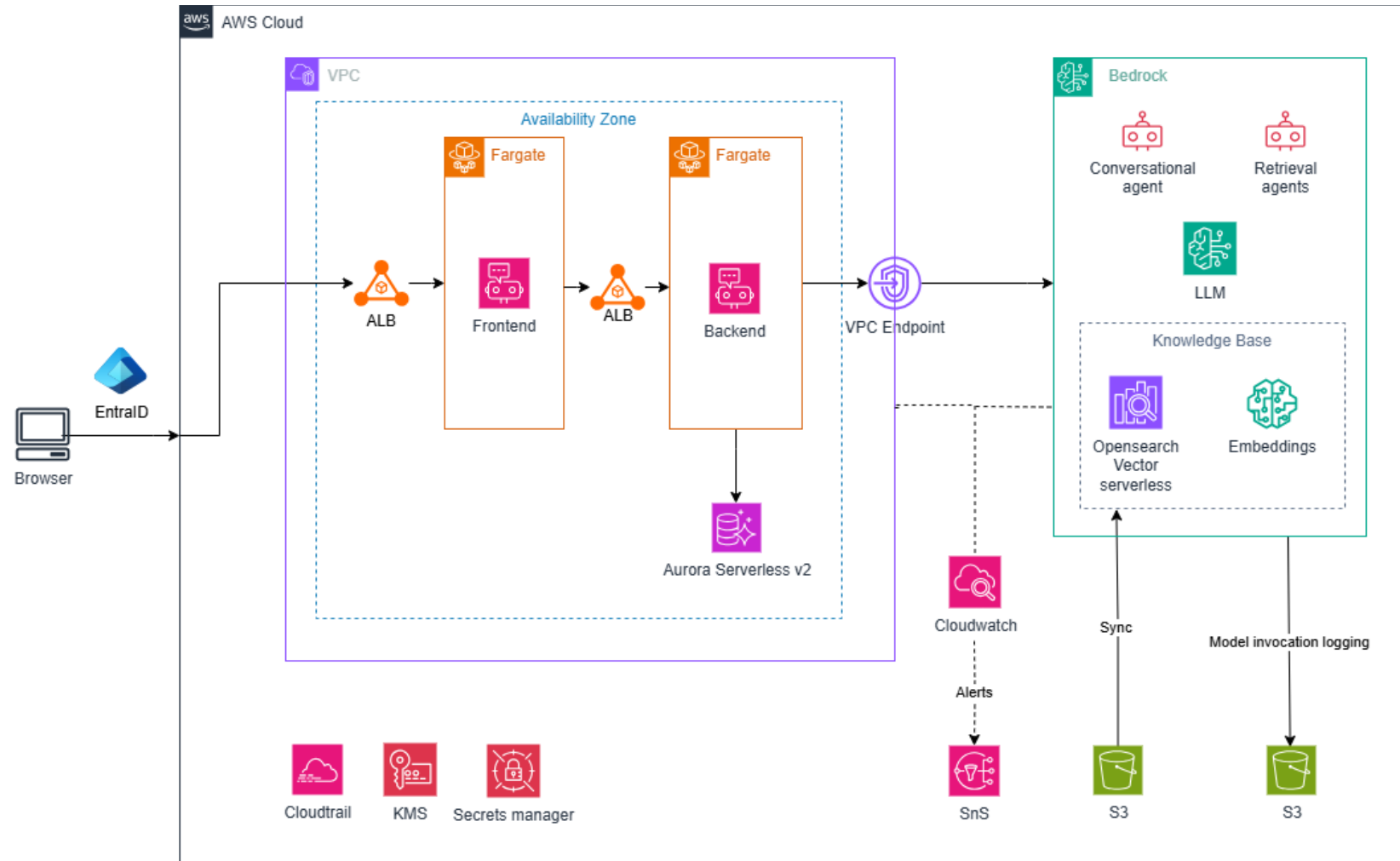
h) Knowledge Bases

- **Purpose:**
Store **domain-specific information** for retrieval.
- **Types:**
 - Compliance Knowledge Base
 - Risk Knowledge Base
 - Forensics Knowledge Base
 - Audit Knowledge Base
- **Fit in the solution:**
Provide structured, trusted content sources for accurate and context-rich answers.

2.3 Flow Summary

1. User submits query → Chatbot Interface → Retrieval/Orchestration Agent.
2. Orchestration Agent decides:
 - Internal KB search: Routes to Agent A → Queries Vector Search Engines → Fetches results from respective Knowledge Bases.
 - Web search: Routes to Agent B → Searches the web.
3. Retrieved content is sent to LLM for synthesis.
4. LLM sends final response back to the Chatbot Interface for the user.

3 AWS Solution Architecture Diagram



3.1 Component Descriptions

3.1.1 Identity & Access

EntralID (Azure AD)

Provides Single Sign-On (SSO) and user authentication. Ensures only authorized employees can access the frontend application.

3.1.2 Frontend Layer (ECS Fargate + ALB)

Application Load Balancer (ALB - external)

Handles incoming traffic from browsers and routes requests to the frontend service.

Frontend (ECS Fargate)

Hosts the user-facing web application (chat interface/dashboard). Runs in containers without managing servers. Scales automatically.

3.1.3 Backend Layer (ECS Fargate + ALB)

Application Load Balancer (ALB - internal)

Balances traffic between frontend and backend microservices for resilience and loose coupling.

Backend (ECS Fargate)

Processes requests from the frontend, orchestrates LLM calls, queries the knowledge base, and returns responses.

3.1.4 Networking

VPC

Provides isolated, secure networking for all components.

VPC Endpoints

Private connections to Bedrock, S3 and other AWS services without traversing the public internet.

3.1.5 AI & Knowledge Layer

Amazon Bedrock

Provides foundation models (LLMs) and managed conversational/retrieval agents. Backend interacts with Bedrock via VPC endpoints.

Conversational Agent: Manages dialogue with users.

Retrieval Agent: Queries external knowledge sources.

LLM: Large language model powering natural language understanding.

Knowledge Base

OpenSearch Serverless (Vector DB): Stores vector embeddings for semantic search and retrieval.

Embeddings Service: Generates embeddings from documents and syncs them into OpenSearch.

3.1.6 Data Management

Amazon Aurora Serverless

Relational database storing structured application and user data and chat history.

Amazon S3

- Stores raw and processed documents for retrieval.
- Receives **Bedrock model invocation logs** for audit and compliance.
- Syncs embeddings with OpenSearch for retrieval augmentation.

3.1.7 Data Integration

- Integration approach is yet to be determined.
- In the interim, documents will be manually uploaded or transferred to S3 until the data integration process is defined.

3.1.8 Observability & Security

KMS (Key Management Service)

Encrypts sensitive data at rest (Aurora, S3, Secrets).

Secrets Manager

Manages secure storage of API keys, DB passwords, and other secrets.

CloudWatch

Collects metrics, logs, and application traces. Triggers alarms on anomalies.

SNS (Simple Notification Service)

Sends alerts (email/SMS) when CloudWatch detects issues.

CloudTrail

Provides governance, compliance, and audit logs of all API activity.

Alerts

- **Fargate (ECS)**
 - **High CPU/Memory:** Alert if CPUUtilization or MemoryUtilization > 80% for sustained period.
 - **Task Failures:** Alert if ServiceRunningTaskCount < ServiceDesiredTaskCount.

SNS Use Case: Notify when tasks crash or scale down unexpectedly.

- **Application Load Balancer (ALB)**

- 5XX Errors: Alert if HTTPCode_ELB_5XX or HTTPCode_Target_5XX spikes.
- Latency: Alert if TargetResponseTime > threshold (e.g., 2s).
- Unhealthy Targets: Alert if HealthyHostCount < expected.

SNS Use Case: Notify on backend failures or degraded performance.

- **Bedrock**

- Invocation Errors: Alert on failed model calls.
- High Latency: Alert if response time exceeds threshold.

SNS Use Case: Notify when AI model calls fail or slow down.

- **Aurora PostgreSQL**

- High CPU/Memory: Alert if CPUUtilization > 80% or FreeableMemory < threshold.
- Connection Saturation: Alert if DatabaseConnections near max.
- Storage Issues: Alert if FreeStorageSpace < 10%.

SNS Use Case: Notify on DB performance degradation or resource exhaustion.

- **S3**

- Access Errors: Alert on spikes in 4xx or 5xx errors.
- Bucket Growth: Alert if BucketSizeBytes grows unexpectedly.

SNS Use Case: Notify on access issues or unexpected data growth.

- **Elasticsearch (OpenSearch)**

- Cluster Health: Alert if ClusterStatus is red or yellow.
- High JVM Pressure: Alert if JVMMemoryPressure > 85%.
- Latency Spikes: Alert if SearchLatency or IndexingLatency > threshold.

SNS Use Case: Notify on search failures or cluster instability.

4 Component Choices

Component	Use	Reason & Pros	Alternatives
Amazon ECS Fargate	Run frontend (web app) and backend microservices in containers without managing servers.	Provides serverless container orchestration with auto-scaling. No EC2 management overhead. Better for long-running workloads than Lambda. Faster to deploy/manage than EKS (no cluster complexity).	- Amazon EC2 (manual container mgmt.) - Amazon EKS (Kubernetes) - AWS Lambda (short-lived tasks)
Amazon Bedrock	Power LLM inference (chatbot, multi-agent orchestration, content analysis).	Provides fully managed access to multiple foundation models (Claude, Mistral, Titan) via API. Eliminates GPU management. Pay-per-use pricing vs costly EC2 GPU hosting. Keeps data within AWS boundary for security & compliance.	- Amazon SageMaker (custom hosting) - Self-hosted EC2 GPU with Autogen - External APIs (OpenAI, Anthropic)
Amazon Aurora Serverless v2	Store structured data (user profiles, metadata, session history).	Provides serverless, auto-scaling relational DB with PostgreSQL/MySQL. Supports ACID transactions unlike DynamoDB. Familiar SQL integration. Lower operational overhead than provisioned RDS. Cost-efficient for spiky workloads.	- Amazon RDS (provisioned) - DynamoDB (NoSQL) - Amazon Neptune (Graph DB)
Amazon OpenSearch Service	Store & query vector embeddings and perform semantic search for RAG.	Provides managed search + vector DB with k-NN. Scales automatically, integrates with IAM/VPC. Supports hybrid search (text + vector). Removes ops burden vs EC2 self-hosted Elasticsearch.	- Aurora pgvector - Pinecone / Weaviate (external) - Self-managed Elasticsearch on EC2

5 Authentication/Authorization

User authentication is handled through Old Mutual's Entra ID, integrated via SAML federation to ensure secure and seamless single sign-on (SSO).

Authorization is implemented through a combination of Active Directory (AD) groups and role-based access controls (RBAC), enforcing the principle of least privilege across all components. Dedicated IAM roles are defined for specific services to isolate permissions and minimize security risks:

- **Bedrock Role** – limited access for invoking and managing model endpoints.
- **Fargate Role** – scoped permissions for containerized application runtime.
- **OpenSearch Role** – restricted to query and index operations as required.
- **Admin Role for Logs** – elevated access to CloudWatch/centralized logging for monitoring and auditing.
- **Deployment Role** – controlled permissions for executing pipeline-driven deployments.

This approach ensures granular access control, auditability, and compliance with enterprise security standards.

6 High availability/Disaster Recovery

- The solution has been architected with all components deployed across multiple Availability Zones (Multi-AZ) to ensure high availability and fault tolerance.
- Furthermore, the platform leverages a serverless architecture, managed through AWS Managed Services (AMS), which provides automated operational support, scalability, and cost optimization. This approach reduces the burden of infrastructure management, allowing the focus to remain on delivering secure, reliable, and resilient business outcomes.

7 Data Lifecycle

The following lifecycle governs data movement and processing within the AI Realm:

1. **Ingestion:** Documents or datasets are ingested via S3 (manual or automated pipeline).
2. **Preprocessing:** Content is cleaned, chunked, and converted into embeddings via Bedrock Embeddings models.
3. **Storage:**
 - **S3** retains raw and pre-processed files.
 - **OpenSearch Vector DB** indexes embeddings for semantic retrieval.
 - **Aurora Serverless** stores metadata, user profiles, and query history.

4. **Retrieval:** The orchestration agent queries OpenSearch for semantically relevant chunks.
5. **Synthesis:** Bedrock LLMs combine retrieved results with user context to generate a grounded response.
6. **Audit & Logging:** CloudTrail and S3 logs capture all model invocation requests and outputs for compliance.

This lifecycle ensures controlled, auditable movement of information across ingestion, enrichment, retrieval, and response layers.

8 Bedrock

8.1 Embedding Models

Embedding models are crucial for RAG as they convert text into numerical vectors that enable fast and accurate retrieval of relevant information from your knowledge base. The following table compares several leading embedding models based on key performance metrics.

Embedding Model	Vector Size	Speed/Cost	Accuracy	Best Use Case
Cohere Embed v3 (Preferred)	384–1024	2/5	5/5	High-quality semantic retrieval
Titan Embeddings G1	1536	4/5	4/5	Large-scale, AWS-native pipelines

Notes on the Models:

- **Cohere Embed v3 (Preferred)** is recognized for its exceptional accuracy, making it the top choice for applications where precision is non-negotiable, such as legal or enterprise knowledge bases. While it may be slower and more expensive, its superior performance often justifies the cost.
- **Titan Embeddings G1** offers a strong balance of speed and accuracy, making it ideal for large-scale projects, especially within existing AWS infrastructures. It provides solid performance without the premium cost of top-tier models.

8.2 Large Language Models (LLMs)

After an embedding model retrieves relevant information, the LLM processes it to generate a final, coherent response. The following table evaluates several LLMs on metrics critical for RAG applications.

Model	Context Length	Hallucination Resistance	Cost-Efficiency	Best Use Case
Claude 3.5 Haiku (Preferred)	200K	4/5	4/5	Fast, cheaper RAG
Claude 3.5 Sonnet	200K	5/5	3/5	Best all-around RAG
Cohere Command R+	128K	4/5	3/5	Precision-grounded KBs
Mistral Mixtral	32K	3/5	5/5	High throughput

Notes on the Models:

- **Claude 3.5 Haiku (Preferred)** is built for speed and affordability. Its rapid response time and improved reasoning make it an excellent choice for real-time applications like interactive chatbots, data extraction, and content moderation where fast, reliable responses are essential.
- **Claude 3.5 Sonnet** is a powerful generalist, excelling in hallucination resistance, which is critical for trustworthy AI applications. Its high performance makes it suitable for a wide range of tasks, particularly when accuracy is paramount.
- **Cohere Command R+** is specifically optimized for advanced Retrieval-Augmented Generation (RAG) workflows and multi-step tool use. Its ability to provide in-line citations and its strong multilingual capabilities make it a reliable choice for enterprise-grade applications where factual accuracy and verifiability are critical.
- **Mistral Mixtral** is ideal for high-throughput scenarios where cost-efficiency and speed are the primary concerns. Its unique architecture allows for quick processing, making it a great option for lightweight knowledge bases or large-scale, high-volume deployments.

8.3 Amazon Bedrock Guardrails

Amazon Bedrock Guardrails provide model-agnostic safety controls that operate on both inputs and outputs to mitigate risks such as prompt attacks, harmful content, and exposure of sensitive information. Guardrails can be versioned, deployed per environment, and enforced across Bedrock, Agents, and Knowledge Bases.

Key Controls

Prompt Injection Protection – Detects and blocks jailbreaks or instruction overrides at both input and output stages.

Content Filtering – Preconfigured categories (hate, insults, violence, sexual, misconduct, prompt attack) with configurable thresholds and block/annotate actions.

Denied Topics & Word Filters – Custom lists and regex patterns to prevent disallowed topics or terms (e.g., financial/medical advice, internal identifiers).

PII Protection – Detects and masks/block personal or sensitive identifiers (e.g., names, credit cards, policy numbers) with custom regex support.

Contextual Grounding – Validates model responses against reference context to reduce hallucinations; can block or re-prompt on low grounding scores.

9 Tech Stack & CI/CD Strategy

Layer	Technology	Rationale	Alternative
Frontend	TypeScript	Type safety, fewer runtime errors, better maintainability & tooling.	JavaScript
	React	Component-based, flexible, large ecosystem & community support.	Angular, Vue
	Tailwind	Utility-first CSS, faster dev, consistent styling, highly customizable.	Bootstrap, Material UI
Backend	Python	Readable, strong ecosystem, excellent for AI/ML & data-heavy apps.	Node.js
	FastAPI	High-performance, async-ready, auto-docs, great for microservices.	Flask, Django

- The solution leverages an Azure DevOps repository as the central source control system, with dedicated CI/CD pipelines configured for both the frontend and backend applications. These

pipelines automate build, test, and deployment processes, ensuring consistency and reducing the risk of manual errors.

- Due to AMS (AWS Managed Services) restrictions, deployments to production are executed manually and require an RFC (Request for Change) approval process, aligning with governance and compliance requirements.

10 Non-Functional Requirements (NFRs)

The AI Realm solution is designed to meet the following enterprise-grade NFRs:

- **Performance & Latency:**
 - <10 seconds median response time for interactive chatbot queries.
 - 95th percentile response time <20 seconds.
- **Availability & Resilience:**
 - 99.9% availability SLA (multi-AZ deployment).
 - Recovery Time Objective (RTO): 1 hour.
 - Recovery Point Objective (RPO): <15 minutes.
- **Scalability:**
 - Support 25 concurrent users in Phase 1 with elastic scale to 50 in Phase 2.
 - Auto-scaling enabled for Fargate tasks and OpenSearch throughput.
- **Security & Compliance:**
 - Aligned with POPIA, FICA, and Old Mutual enterprise security controls.
 - All data encrypted in transit (TLS 1.2+) and at rest (KMS).
 - IAM enforced least-privilege access with role separation.
- **Auditability:**
 - All queries and model invocations logged in S3.
 - Audit-ready evidence for regulatory reporting.
- **Cost Efficiency:**
 - Monthly operating cost ceiling: \$(TBC) during Phase 1.
 - Guardrails to scale down compute in off-peak windows.

11 AWS Cost Assessment

Costs indicated below are based on resource costs in AWS.

The calculations were done on the AWS pricing calculator, view detailed cost breakdown:

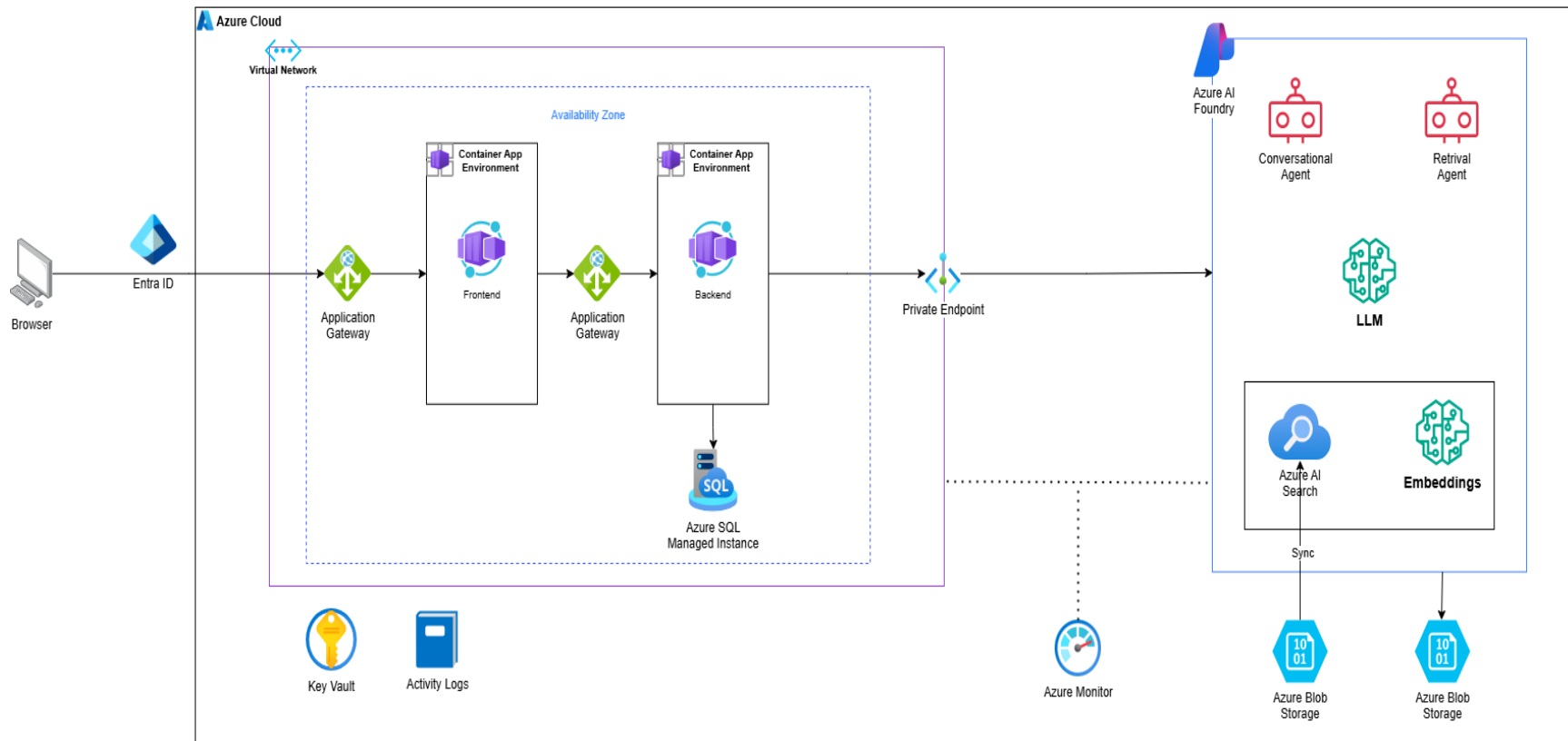
<https://calculator.aws/#/estimate?id=466943d007564ff12de9d770bcd0728f842798e2>

Region: EU (Ireland)

Currency: USD

Service	Description	Monthly	First 12 months total
Amazon Aurora PostgreSQL		\$54.56	\$654.72
Amazon CloudWatch		\$0.00	\$0.00
Amazon OpenSearch Service	Vectorstore	\$368.86	\$4 426.32
AWS Bedrock	Embeddings	\$1.00	\$12.00
AWS Bedrock	LLM	\$756.00	\$9 072.00
AWS CloudTrail		\$0.00	\$0.00
AWS Fargate	Frontend	\$48.22	\$578.64
AWS Fargate	Backend	\$82.02	\$984.24
AWS Key Management Service		\$9.00	\$108.00
AWS Secrets Manager		\$5.40	\$64.80
Data Transfer		\$0.00	\$0.00
S3 Standard		\$7.31	\$87.72
Amazon Simple Notification Service (SNS)		\$0.00	\$0.00
		\$1 332.37	\$15 988.44

12 AZURE Solution Architecture Diagram



12.1 Component Description

12.1.1 Identity & Access

EntraID (Azure AD)

Provides Single Sign-On (SSO) and user authentication. Ensures only authorized employees can access the frontend application.

12.1.2 Frontend Layer

Application Gateway (Load Balancer)

Handles incoming traffic from browsers and routes requests to the frontend service and secures it using Web Application Firewall (WAF) protection.

Frontend (Container App Environment)

Hosts the user-facing web application (chat interface/dashboard). Runs in containers without managing servers. Scales automatically using user-defined rules.

12.1.3 Backend Layer

Application Gateway (Load Balancer)

Handles incoming traffic from frontend to backend container app environment. It uses intelligent routing, SSL offloading while also using WAF protection.

Backend (Container App Environment)

Processes requests from the frontend, orchestrates LLM calls, queries the knowledge base, and returns responses.

12.1.4 Networking

Virtual Network

Isolated network in the Azure cloud environment for all components

Private Endpoint

Processes requests to Azure AI Studio, Azure Blob, and other services securely through the private network.

12.1.5 AI & Knowledge Layer

Azure AI Foundry

Provides foundation models (LLMs) and managed conversational/retrieval agents. Backend interacts with Azure AI Foundry via Private endpoints.

Conversational Agent: Manages dialogue with users.

Retrieval Agent: Queries external knowledge sources.

LLM: Large language model powering natural language understanding.

Knowledge Base

Azure AI Search (Vector DB): Stores vector embeddings for semantic search and retrieval.

Embeddings Service: Generates embeddings from documents and syncs them into Azure AI Search.

12.1.6 Data Management

Azure SQL Managed Instances

A SQL relational database, managed and secured by Azure for storing structured application and user data and chat history.

Azure Blob

- Stores raw and processed documents for retrieval.
- Receives **Azure AI Foundry invocation logs** for audit and compliance.
- Syncs embeddings with OpenSearch for retrieval augmentation.

12.1.7 Observability & Security

Key Vault

Encrypts and secures keys and passwords used in the ecosystem.

Activity Logs

Logs all users and actions made on the environment.

Azure Monitor

Monitors all components and alerts can be created using defined rules for each component within their respective measurable metrics.

Azure Defender for Cloud

Recommended by Microsoft to secure generative AI environments within AI Foundry.

12.2 Old Mutual Availability of Azure Services

OM has provided iOCO with a comprehensive list of the services OM use or have access to in the AWS accounts along with their Azure equivalents. The colour codes used are as follows: Green indicates the service is available in Azure, and Yellow means that Adam and his team are still working on making it available.

Note that some services used in the Azure architecture is in yellow in the list below.

- API Gateway – Application gateway/ Front door
- Athena – kind of like Synapse
- CloudFormation – Terraform
- CloudTrail – Activity Logs
- CloudWatch – Azure Monitor
- CloudWatch Events - Azure Monitor
- CodeBuild – Azure Devops
- CodeCommit - Azure Devops
- CodePipeline- Azure Devops
- Compute Optimizer – Azure Advisor
- Config – Sort of like Azure Policies
- Cost Explorer – Cost Management
- Data Transfer – this might just be the network cost item?
- Direct Connect – Express Route
- Directory Service – Entra ID
- DynamoDB – Cosmos DB
- EC2 Container Registry (ECR) – Azure Container Registry
- Elastic Compute Cloud (EC2) – Virtual Machines
- Elastic Container Service (ECS) – Azure Container Instances/ Container Apps/ AKS
- Elastic Load Balancing – Azure Load balancer / Application gateway
- Glue – Data Factory
- GuardDuty - Sentinel
- Key Management Service (KMS) – Key Vault
- Kinesis Firehose – Azure Stream Analytics

- Lambda - Functions
- Macie – Azure Security Centre
- Relational Database Service (RDS) – SQLMI/ Postgresql Flex / Mysql Flex
- Route 53 – Azure DNS
- Secrets Manager – Azure Key vault
- Security Hub - Azure Security Center with Azure Defender
- Service Catalog – Managed Applications
- Simple Notification Service (SNS) – Azure notification hub
- Simple Queue Service (SQS) – Azure queue storage
- Step Functions – Logic Apps
- Systems Manager – Azure Automation & Update Management
- Virtual Private Cloud (VPC) – Virtual Network
- X-Ray – Application Insights

13 AZURE Cost Assessment

DISCLAIMER: These prices as of 29/8/2025 for UK South Region. Some services costs may vary depending on the usage/size/model

Service	Size	Cost (Pay as you go)	Cost (Per Month)
Azure Container Apps	4 vCPU 8 GiB Mem	First 180,000 vCPU per second, 360,000 GiB/s, and 2 million requests each month are free. Then \$0.00000400 Per vCPU/sec \$0.00000400Per GiB/sec	Assuming two idle instances (1 for backend 1 for frontend): \$252.29
Virtual Network	100 GB Data Transfer Outbound 100 GB Data Transfer Inbound	\$0.01 per GB	\$2.00
Application Gateway	Web Application Firewall v2 744 hours	\$0.45 per Hour	\$356.23 (2 Compute Units)

Monitoring	1 GB Ingestion Per Day 3 Month Monitoring Data Retention 100 SMS per month for SA number	Varies	\$137.60
Key Vault	Hardware Security Module Protected Keys and Certificate Renewal	\$1.00 per HSM key \$3.00 per Certificate \$0.03 per 10000 operations	\$4.03
Azure AI Search	Standard S1	\$0.34/hour	\$249.98
Azure AI Foundry	Model: o4-mini 2025-04-16 Global	Input: \$1.10 Cached Input: N/A Output: \$4.40 Per Million Token	N/A
Azure Blob Storage	Access Tier Hot 3 TB	N/A	\$58.83
Estimated total monthly cost for above services			\$1,063.37